rainmaker

_travelers guide
_for creating microclimates

16 delay taps | **64** comb taps | **128** presets

› then noon = 12 o'clock tags = the panel drawing

stereo 16-tap rhythm delay + 64-tap comb resonator · every tap has its own filter, pan and pitch · 128 presets

01 what it is

rm_01

two effects	a 16-tap stereo rhythm delay + a 64-tap comb resonator · 96 kHz / 32-bit · 128 presets
every tap	its own level, pan, filter (lp · bp · hp), pitch shift –16 … +15 and detune
the comb	long = echoes · short + high feedback = karplus-strong plucks and pitched ringing

02 two ways to edit

rm_02 · the big idea

[edit/tap#]	flips the 16 blue buttons: green `edit` LED = edit pages · `tap#` LED = pick taps 1 – 16
› tap#	a white tap edit button, a tap button, turn = that one tap
› edit	a white tap edit button + push the encoder = set all 16 taps the same
push twice	in edit: cycle 8 ramp shapes across the taps (swells, decays, filter sweeps)
[mute]	shows levels · tap buttons toggle mutes (crosshatched bars)

03 edit pages · top row

rm_03 · edit LED on

[rand.]	turn = what gets randomized (15 targets + random) · push = roll it now
[grid]	taps per beat 1 2 3 4 6 8 12 16 · tap n lands at n ÷ grid beats
[piles]	stacks: taps in a stack share the last one's time · pitch them into chords
[time] [div]	beat time in ms + bpm · clock in divide /1 – /256
[wave] [level] [rate]	delay time lfo: off · sine · tri · vibrato 1 · 2 · down/up chirp · random · 0 – 15 (10 ms) · Hz

04 first flight

rm_04 · a clean start

1. [load] › preset 1 (1-tap) › push
2. sound into `in L` · `out L` `out R` out
3. `mix` to noon · tap [tempo] three times
4. raise delay feedback for repeats
5. [edit/tap#] to taps › [mute] › light more taps
6. save it before experimenting

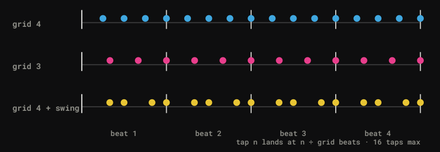
05 edit pages · bottom row

rm_05 · fig.01

[qty] [size]	grains 1 · 2 · 4 · 1 unshaped · 5 – 671 ms
[tap#]	feedback tap L / R: 1 – 16, all (no pitch shift) or comb size · press again for R only
[slip]	nudge the feedback time –16 … +15 · rushing or dragging echoes
[pitch]	feedback pitch + detune, L / R · press to step through the four values
[type] [amt]	16 grooves (swing, waltz, accelerando …) · 0 – 100 % blend
[reverse]	turn = reverse on · press again = the delay's own wet/dry

_fig.01 : where the taps land

tap grid · groove



06 the taps

rm_06 · tap edit buttons

[level]	0 – 15, exponential · 0 = off · lit tap LED = level above 0
[pan]	7L … 0 … 7R · balance in stereo, pan in mono
[cut] [q] [type]	cutoff 0 – 4095 (≈ Hz) · q 0 – 15 · none · lp · bp · hp
[pitch] ×2	–16 … +15 semitones · press again = detune (under ½ semitone)

07 the comb

rm_07 · comb pages

[patt+slope]	16 tap patterns (fibonacci, early, late, ess …) · again = slope: flat, rising, falling, rise/fall
[taps+fb]	1 – 64 taps · again = feedback: guitar · sitar · clarinet (cleanest self-osc) · raw
[clk src]	size from comb clk · pot + cv + 1v/oct · encoder · delay time · [clk div] [size] set divide / ms
`1v/oct`	+1 V halves the size · no attenuator · [wave] [level] [rate] = size lfo

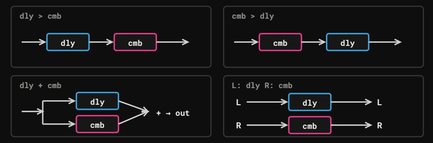
08 routing+config

rm_08 · press 1 · 2 · 3 · fig.02

1 routing	dly › cmb · cmb › dly · dly + cmb · L = dly, R = cmb
2 input	stereo · mono L+R · mono L
3 trigger	what [trigger] and `trig` do · freeze / reverse state shown

_fig.02 : routing

routing+config, first press



09 trigger modes

rm_09 · routing+config ×3

freeze · gated freeze — toggle · hold
loops the last feedback-tap length · taps keep echoing it

reverse · gated reverse — toggle · hold
plays the buffer backwards

ping input · shaped pulse — excite
noise burst or soft pulse into the input · pluck the comb

randomize — per rand. page
rolls whatever [rand.] is set to

mutes · bypass — dly · cmb · all
toggle a section's mute or bypass

pingpong · preset load · clear — utility
swap L/R feedback · load the selected preset · wipe both buffers

10 mod a · mod b

rm_10 · white button, turn, push

pick	press its white button · turn · push to lock ("?" = not locked yet)
offsets	tap filter cut · q · balance
sets	grain size · fb pitch / detune / tap# · lfo wave, level, rate · groove · wet/dry
also	comb density · pattern · stereo spread · all tap pitch · grid · clk div · preset #
mod b only	rhythm delay time · mod A is normalled to mod B

11 the jacks

rm_11 · cv adds to its knob

delay	`feedback` `pitch shift` `tone` · `trig` · `clk` sets the beat (× div)
comb	`size` `feedback` · `1v/oct` · `clk` when clk src = comb clk

12 quiet? distorted?

rm_12 · meters+gain

[meters+gain]	8 meters: in · out · dly in/fbk/out · cmb in/fbk/out · press = dly or cmb gain
gains	about 8 – 10 · lower dly as taps pile up
inputs	attenuators near noon · keep IN off the top
two mixes	`mix` pot = analog dry/wet · [reverse] page = inside the delay

13 presets

rm_13 · 128 slots

save — save › turn › push
save again = edit the 10-character name · "saved!"

load — load › turn › push
knob values hold until each knob moves · hold load + push = active effects only

sysex — save + load together
usb on the back · dump or send presets over midi sysex

try — 38 · 66 · 68 · 82+
karplus-strong · modal ring · delay*size · richard devine's set · 70 – 81 empty

14 hidden moves

rm_14

push + turn	coarse steps on time, rate, cut, groove amount
mod pages	push the encoder = lfo level ×⅛
[tap#] / [slip]	push the encoder = ping-pong (crossed arrows on screen)
[clear]	wipes both delay memories · runaway feedback fix
pitch knob	dead band at noon · ±2 oct, taps only

_fig.03 : the panel

numbers match the legend



`edit` **edit/tap#** · pages ↔ per-tap
`pages / taps` **16 blue buttons** · edit pages · or taps 1 – 16
`tap edit` **white row** · mute · level · pan · cut · q · type · pitch
`enc` **encoder** · turn · push · push+turn = coarse
`pots` **8 large** · feedback · pitch · tone · size · comb fb · mix · mod a · mod b
`cv` **cv jacks** · 0 – 10 V · small knobs scale them
`trig` **trigger** · big red button + jack
`tempo` **tempo** · tap 3× · clk in overrides
`presets` **red column** · routing · meters · save · load
`mod` **mod a** · **b** · button = pick destination

15 specs

rm_15

audio	±5 V in and out · in L feeds in R
cv · clock	cv 0 – 10 V · clk in / out 0 – 5 V · clk out = tempo LED
size	36 hp · 44 mm deep
power	270 mA +12 V (nearly 300 at start) · 24 mA –12 V